

Model of Communications System

Source $\rightarrow$ Transmitter $\rightarrow$ Transmission System $\rightarrow$ Receiver
 $\downarrow$
Destination

Data $\xrightarrow{\text{encoded}}$ Signal $\rightarrow$ flows through transmission system

Encoding

$\rightarrow$ Encoding means changing a signal's properties, (like voltage, frequency, phase) to represent data.

$\rightarrow$ The more distinct states you can use, the more data you can transmit.

Transmission Signal :-

It can be an electro-magnetic wave or electric signal.

EM wave example:-

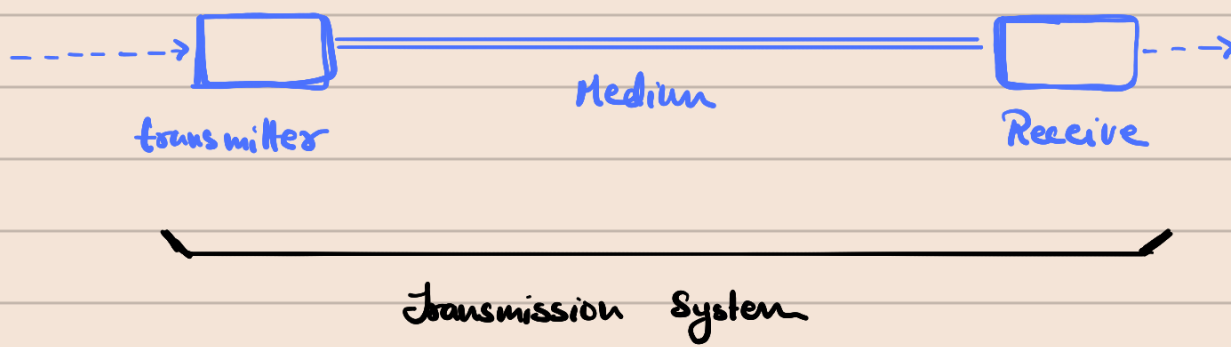
Radio Wave, Light wave, Microwave

Electrical Signal Example:-

Alternating Current, Pulses

* The simplest form of signal (both in EM & Electrical) is Sine Wave.

Transmission System = Transmitter + medium + Receiver



Medium

→ Guided:

Signal travel through a physical path
ex: Electrical Cables, fibre optic Cable.

→ Unguided:

Signal travel through open space without path
ex: E.M. Waves, example Radiowave or light.

Successful Transmission of Data:

Depends upon two factors:-

1. Signal Quality.
2. Medium Quality.

1. Signal

Properties (or Characteristics)

1) Continuous or Discrete :-

Continuous : No break or discontinuity within signal. No Gaps.

example :

Speech signal is continuous in both time and amplitude aspect.

Discrete : Contains finite number of discrete values which can be represented.

Example :-

Binary Data :-

Time Aspect :

Signal is only available at
 $t=1, t=2, t=3$

Not at $t=1.2$ or $t=2.3$

Amplitude Aspect :-

Signal can only have 1 or 0
no 0.1 or 0.3 values

2. Periodic And Aperiodic

A periodic signal repeats its pattern after a fixed interval of time, called a period.

Sine Wave Characteristics

Equation :-

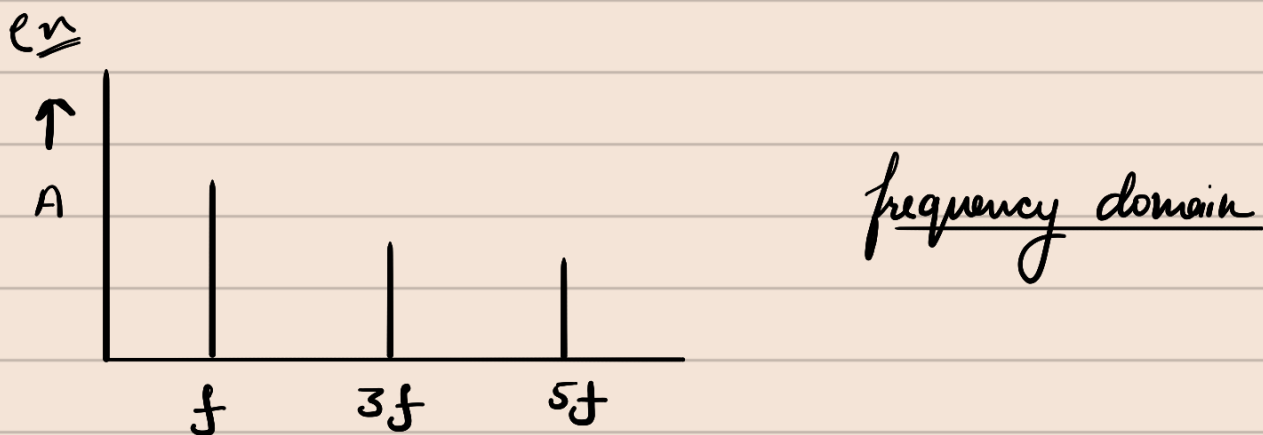
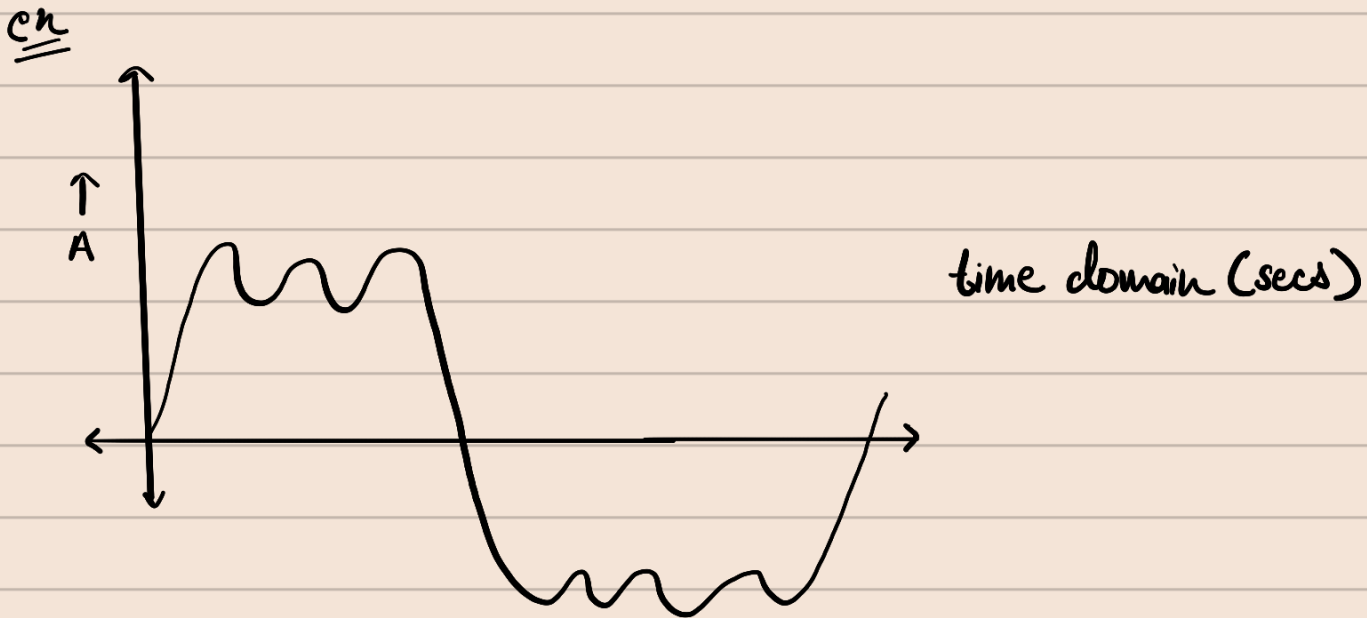
$$s(t) = A \sin(2\pi \cdot f t + \phi)$$

A = amplitude

f = frequency = $\frac{1}{T}$ (time period)

ϕ = phase

Representation of Signals:-



fun fact ✓ these frequencies are harmonic.

Note: frequencies are harmonic if you can find a single fundamental f such that all frequencies = $n \cdot f$.

where n is an integer.

Fourier Analysis

It states that a periodic signal can be expressed as a sum of sine waves with different frequencies, amplitudes and phases.

Sinusoidal Component

The effect of transmission media on a signal can be analysed by examining the effects on these components.

Signalling Concepts

- 1) Spectrum : Range of frequencies contained in a signal: on f to $5f$
- 2) Absolute Band width: width of spectrum : ex $5f - f = 4f$
- 3) Effective Bandwidth : Portion of spectrum where most of the useful transmission happens.
- 4) D.C. Component : the component that determines the overall average of the wave.

$$\text{ex: } x(t) = \underline{3} + \sin(\omega t)$$

$\uparrow$
D.C. offset.

Square Wave :-

$$s(t) = A \sum_{\substack{k=1 \\ \dots \text{odd}}}^{\infty} \frac{1}{k} \sin(2\pi \cdot k f t)$$

System Bandwidth

* System bandwidth is the difference between the highest and the lowest frequency that a system can transmit effectively.

- ↳ Signal outside the range are not completely blocked.
- ↳ they are heavily distorted
- ↳ so they become useless for transmission.

* System Bandwidth is constrained by:-

- the physical behavior of its components
- and fundamental properties of matter and energy.

Relationship between Data Rate And Bandwidth:-

Bandwidth = Range of frequencies a transmission system can transmit effectively, measured in hertz.



Large Bandwidth allows the system to support faster change in signal

↑ faster signal enables higher Data Rates

∴ data rate $\propto$ Bandwidth

Example :-

Bits: 0 1 0 0 0 0 1 0 0

Pulses before transmission:

Bit rate: 2000 bits per second

Pulses after transmission:

Bandwidth 500 Hz

Bandwidth 900 Hz

Bandwidth 1300 Hz

Bandwidth 1700 Hz

Bandwidth 2500 Hz

Bandwidth 4000 Hz

Data: 010000100

→ ideal digital signal

Same signal after passing through system with limited bandwidth

higher bandwidth
= more details are preserved

Low bandwidth

→ Attenuate higher frequencies

Lower frequencies remain

→ They look like smooth sinewave.

* A system with limited bandwidth cannot pass high frequency components effectively.

* To preserve the shape of the signal, the rate at which the signal changes must be reduced so that its frequency contents fits within the system bandwidth.

how many frequency components fit within bandwidth

Data Rate (How fast bits are sent)

Time per bit

Base frequency of signal

Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

* Increasing data Rate pushes the signal beyond the system's bandwidth, stripping away harmonics and destroying wave form shapes.

higher bandwidth of System $\approx$ Greater Bandwidth of System

higher the speed (frequency) of Signal $\approx$ Greater bandwidth of signal
 $\approx$ More data can be transmitted

2 Data And Signals

Data $\rightsquigarrow$ Signal $\rightsquigarrow$ Transmission System

→ These are three distinct entities.

→ Each can be classified as either analogue or digital.

1. Data $\rightsquigarrow$ Analogue

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Digital

Take on continuous values on some intervals, like Voice temp, pressure.

Take on discrete values

ex:- integers / cents

2. Signal :- $\rightsquigarrow$ Analogue

Digital

Continuously varying E. M. wave,
propagating over a transmission medium.

Sequence of Discrete, discontinuous
voltage pulses (representing data)

3.7 Transmission:-

Data transmission is the communication of data by propagation and processing of signals.

Analog Data conveyed by Digital Signal uses CODEC

Digital Data conveyed by Analog signal uses MODEM.